MTH 346/646 - 17th - 10/23

Fun fact: The elliptic curve $E : y^2 = x^3 - 37x^2 + 36x$ has infinitely many rational points on it. For any rational point $P = (x, y) \in E(\mathbb{Q})$, the number x is the square of a rational number.

Last time: Proof of Lemma 3.

Overview of Lemma 4

Step 1:

We will only prove Lemma 4 for an elliptic curve with a rational point of order 2. If we move the point of order 2 to $T = (0, 0)$, the curve takes the form $C : y^2 = x^3 + ax^2 + bx$. This curve has a "friend"

$$\overline{C} : y^2 = x^3 + \overline{a}x^2 + \overline{b}x,$$

where $\overline{a} = -2a$ and $\overline{b} = a^2 - 4b$.

There are morphisms $\phi : C \to \overline{C}$ and $\overline{\phi} : \overline{C} \to C$ so that for any $P \in C(\mathbb{Q})$,

$$\overline{\phi}(\phi(P)) = 2P$$

and for any $\overline{P} \in \overline{C}(\mathbb{Q})$,

$$\phi(\overline{\phi}(\overline{P})) = 2\overline{P}.$$

This means that a point $P \in C(\mathbb{Q})$ is twice a point (that is, in $2C(\mathbb{Q})$) if and only if it can be written in the form $P = \overline{\phi}(Q)$ for some $Q \in \overline{C}(\mathbb{Q})$ *and* Q can be written in the form $Q = \phi(R)$ for some $R \in C(\mathbb{Q})$.

Q1: What are the kernels of ϕ and $\overline{\phi}$?

A: $\ker \phi = \{\mathcal{O}, T\}$ and $\ker \overline{\phi} = \{\overline{\mathcal{O}}, \overline{T}\}$.

Q2: What are the images of ϕ and $\overline{\phi}$?

A: The image of $\overline{\phi}$ is the set of points in $C(\mathbb{Q})$ for which the x-coordinate is the square of a nonzero rational number (most of time T is not in the image of ϕ, but it is when b is a square). Same thing with ϕ.

Step 2: Two more homomorphisms.

Let $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ be the set of nonzero rational numbers under multiplication mod squares. There is a homomorphism $\alpha : C(\mathbb{Q}) \to (\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$ defined by taking $\alpha((x, y)) = x \cdot (\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$.

Similarly, there is a homomorphism $\overline{\alpha} : \overline{C}(\mathbb{Q}) \to (\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$.

From the answers above, we have that $\ker \alpha = \text{im } \overline{\phi}$ and $\ker \overline{\alpha} = \text{im } \phi$.

Theorem: The image of α is a finite subgroup of $(\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$, similarly, the image of $\overline{\alpha}$ is a finite subgroup of $(\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$.

Calculation: $[C(\mathbb{Q}) : 2C(\mathbb{Q})] \leq |\text{im } \alpha| \cdot |\text{im } \overline{\alpha}|$.

Proof of Lemma 4 - Start of Step 1

If $C : y^2 = x^3 + ax^2 + bx$, then the discriminant of C is $b^2(a^2 - 4b)$. We assume that $a^2 - 4b \neq 0$.
We will construct a map from C to $\overline{C} : y^2 = x^3 + \overline{a}x^2 + \overline{b}x$, where $\overline{a} = -2a$ and $\overline{b} = a^2 - 4b$.
We could then do this process again, and get a curve $\overline{\overline{C}} : y^2 = x^3 + \overline{\overline{a}}x^2 + \overline{\overline{b}}$.
We have $\overline{\overline{a}} = -2\overline{a} = -2(-2a) = 4a$ and $\overline{\overline{b}} = \overline{a}^2 - 4\overline{b} = (-2a)^2 - 4(a^2 - 4b) = 4a^2 - 4a^2 + 16b = 16b$.
Thus, $\overline{\overline{C}} : y^2 = x^3 + 4ax^2 + 16bx$. If we set $y = 8Y$ and $x = 4X$, we get

$$64Y^2 = 64X^3 + 64aX^2 + 64bX,$$

and dividing we get $Y^2 = X^3 + aX^2 + bX$, so $\overline{\overline{C}} \cong C$.
The map $\phi : C \to \overline{C}$ is given by $\phi(x, y) = (\overline{x}, \overline{y})$, where

$$\begin{aligned} \overline{x} &= x + a + \frac{b}{x} = \frac{y^2}{x^2} \\ \overline{y} &= y\left(\frac{x^2 - b}{x^2}\right). \end{aligned}$$

We calculate that

$$\begin{aligned} \overline{x}^3 + \overline{a}\overline{x}^2 + \overline{b}\overline{x} &= \overline{x}\left(\overline{x}^2 - 2a\overline{x} + (a^2 - 4b)\right) \\ &= \frac{y^2}{x^2}\left(\frac{y^4}{x^4} - 2a\frac{y^2}{x^2} + (a^2 - 4b)\right) \\ &= \frac{y^2}{x^2}\left(\frac{y^4 - 2ax^2y^2 + a^2x^4 - 4bx^4}{x^4}\right) \\ &= \frac{y^2}{x^2}\left(\frac{(y^2 - ax^2)^2 - 4bx^4}{x^4}\right) \\ &= \frac{y^2}{x^6}\left((x^3 + bx)^2 - 4bx^4\right) \\ &= \frac{y^2}{x^6}\left(x^6 + 2bx^4 + b^2x^2 - 4bx^4\right) \\ &= \frac{y^2}{x^6}\left(x^6 - 2bx^4 + b^2x^2\right) \\ &= \frac{y^2}{x^6}\left(x^3 - bx\right)^2 \\ &= \frac{(y(x^2 - b))^2}{x^4} = \overline{y}^2. \end{aligned}$$

Proposition: Assume the notation above.

(a) The map

$$\phi : C \to \overline{C}$$

given by

$$\phi(P) = \begin{cases} \left(\frac{y^2}{x^2}, \frac{y(x^2-b)}{x^2}\right) & \text{if } P \neq \mathcal{O}, T \\ \overline{\mathcal{O}} & \text{if } P = \mathcal{O} \text{ or } T \end{cases}$$

is a homomorphism. The kernel of $\phi(P)$ is $\{\mathcal{O}, T\}$.

(b) Applying the same process gives a map $\overline{\phi} : \overline{C} \to \overline{\overline{C}} \cong C$. We get a map $\psi : \overline{C} \to C$ given by

$$\psi(P) = \begin{cases} \left(\frac{\overline{y}^2}{4\overline{x}^2}, \frac{\overline{y}(\overline{x}^2-\bar{b})}{8\overline{x}^2}\right) & \text{if } \overline{P} = (\overline{x}, \overline{y}) \neq \overline{\mathcal{O}}, \overline{T} \\ \mathcal{O} & \text{if } \overline{P} = \overline{\mathcal{O}} \text{ or } \overline{T}. \end{cases}$$

The composition $\psi \circ \phi : C \to C$ is multiplication by two: $\psi \circ \phi(P) = 2P$. Also, $\phi \circ \psi : \overline{C} \to \overline{C}$ is multiplication by 2.

Proof:

(a) It is easy to see that if ϕ is a homomorphism, then its kernel is $\{\mathcal{O}, T\}$. It suffices to show that it is a homomorphism. We need to show that $\phi(P_1 + P_2) = \phi(P_1) + \phi(P_2)$.

Case I: $P_1 = P = (x, y)$ and $P_2 = T = (0, 0)$.

Fact: $P + T = \left(\frac{b}{x}, -\frac{by}{x^2}\right)$. (Do you want to see a proof?)

Let $\phi(P + T) = (\overline{x}(P + T), \overline{y}(P + T))$. We get

$$\begin{aligned} \overline{x}(P + T) &= \left(\frac{y(P + T)^2}{x(P + T)^2}\right) \\ &= \frac{(-by/x^2)^2}{(b/x)^2} = \frac{b^2y^2}{b^2x^2} = \frac{y^2}{x^2} = \overline{x}(P). \end{aligned}$$

We have

$$\begin{aligned} \overline{y}(P + T) &= \frac{y(P + T)(x(P + T)^2 - b)}{x(P + T)^2} \\ &= \frac{(-by/x^2)((b/x)^2 - b)}{(b/x)^2} \\ &= \frac{(-by/x^2)(b^2 - bx^2)}{b^2} \\ &= (y/x^2)(x^2 - b) = \overline{y}(P). \end{aligned}$$

This proves that $\phi(P + T) = \phi(P) + \phi(T) = \phi(P)$.

If $P = (x, y)$, then

$$\phi(-P) = \phi((x, -y)) = \left(\frac{(-y)^2}{x^2}, \frac{(-y)(x^2 - b)}{x^2}\right) = -\phi(P).$$

Case II: Finally, we will prove that if $P_1 + P_2 + P_3 = \mathcal{O}$, then $\phi(P_1) + \phi(P_2) + \phi(P_3) = \overline{\mathcal{O}}$. This will prove the statement because then $\phi(P_1) + \phi(P_2) = -\phi(P_3) = \phi(-P_3) = \phi(P_1 + P_2)$. Because we handled the case of $P + T$, we can also assume that P_1, P_2 and P_3 are all different from T.

Let $y = \lambda x + \nu$ be the line that goes through P_1, P_2 and P_3 and note that $\nu \neq 0$.

Claim: $\phi(P_1)$, $\phi(P_2)$ and $\phi(P_3)$ are all on the line $y = \overline{\lambda} x + \overline{\nu}$, where

$$\overline{\lambda} = \frac{\nu\lambda - b}{\nu}, \quad \overline{\nu} = \frac{\nu^2 - a\nu\lambda + b\lambda^2}{\nu}.$$

Proof: Let $P_1 = (x_1, y_1)$. We get

$$\begin{aligned}
\overline{\lambda}\overline{x}_1 + \overline{\nu} &= \frac{\nu\lambda - b}{\nu}\left(\frac{y_1^2}{x_1^2}\right) + \frac{\nu^2 - a\nu\lambda + b\lambda^2}{\nu} \\
&= \frac{(\nu\lambda - b)y_1^2 + (\nu^2 - a\nu\lambda + b\lambda^2)x_1^2}{\nu x_1^2} \\
&= \frac{\nu\lambda(y_1^2 - ax_1^2) - by_1^2 + b\lambda^2 x_1^2 + \nu^2 x_1^2}{\nu x_1^2} \\
&= \frac{\nu\lambda(y_1^2 - ax_1^2) - b(y_1 - \lambda x_1)(y_1 + \lambda x_1) + \nu^2 x_1^2}{\nu x_1^2}.
\end{aligned}$$

We have $y_1^2 = x_1^3 + ax_1^2 + bx_1$ and we also have $y_1 - \lambda x_1 = \nu$. Using this we get

$$\begin{aligned}
&= \frac{\lambda(x_1^3 + bx_1) - b(y_1 + \lambda x_1) + \nu x_1^2}{x_1^2} \\
&= \frac{\lambda(x_1^3 + bx_1) - by_1 - b\lambda x_1 + \nu x_1^2}{x_1^2} \\
&= \frac{\lambda x_1^3 + \nu x_1^2 - by_1}{x_1^2} \\
&= \frac{x_1^2(\lambda x_1 + \nu) - by_1}{x_1^2} \\
&= \frac{x_1^2 y_1 - by_1}{x_1^2} \\
&= \frac{y_1(x_1^2 - b)}{x_1^2} \\
&= \overline{y}_1.
\end{aligned}$$

This calculation actually shows that any point on the line $y = \lambda x + \nu$ and the curve $y^2 = x^3 + ax^2 + bx$ gets mapped under ϕ to a point on the line $y = \overline{\lambda} x + \overline{\nu}$.

(Strictly speaking this is not quite enough. We need to know that in the case that $\phi(P_1)$, $\phi(P_2)$ and $\phi(P_3)$ are not all distinct, we get the "right" point on this line, but that follows from continuity.)

(b) We want to show that $\psi \circ \phi(P) = 2P$. A calculation shows that

$$2P = \left(\frac{(x^2 - b)^2}{4y^2}, \frac{(x^2 - b)(x^4 + 2ax^3 + 6bx^2 + 2abx + b^2)}{8y^3} \right)$$

We compute that

$$\psi \circ \phi(x, y) = \psi \left(\frac{y^2}{x^2}, \frac{y(x^2 - b)}{x^2} \right)$$

$$= \left(\frac{\left(\frac{y(x^2-b)}{x^2}\right)^2}{4\left(\frac{y^2}{x^2}\right)^2}, \frac{\frac{y(x^2-b)}{x^2} \cdot \left(\left(\frac{y^2}{x^2}\right)^2 - (a^2-4b)\right)}{8\left(\frac{y^2}{x^2}\right)^2} \right)$$
$$= \left(\frac{(x^2-b)^2}{4y^2}, \frac{\frac{y(x^2-b)}{x^2} \cdot (y^4 - (a^2-4b)x^4)}{8y^4} \right)$$
$$= \left(\frac{(x^2-b)^2}{4y^2}, \frac{\frac{(x^2-b)}{x^2} \cdot (x^2(x^2+ax+b)^2 - (a^2-4b)x^4)}{8y^3} \right)$$
$$= \left(\frac{(x^2-b)^2}{4y^2}, \frac{(x^2-b)\left[(x^2+ax+b)^2 - (a^2-4b)x^2\right]}{8y^3} \right)$$
$$= \left(\frac{(x^2-b)^2}{4y^2}, \frac{(x^2-b)\left(x^4+2ax^3+6bx^2+2abx+b^2\right)}{8y^3} \right).$$

This proves the desired statement (at least when x and y are both nonzero). It takes a little bit more work to prove it when x or y is zero.

Part (b) of the proposition follows from part (a), just applying it to $\overline{C}$ and not C.

QED Proposition

What's going on here?

If C is a curve, and H is a group of automorphisms of the curve C, it is possible to create a "quotient curve" C/H. There is a quotient map $\phi : C \to C/H$ with degree equal to the order of H. This map has the property that if P_1 and P_2 are points on C for which there is an automorphism $h \in H$ so that $h(P_1) = P_2$, then $\phi(P_1) = \phi(P_2)$.

In our case, $C : y^2 = x^3 + ax^2 + bx$ and H is a set with two automorphisms: the identity automorphism, and the map $h : C \to C$ given by $h(P) = P + T$.